Theoretical Minimum Examination: Paper 001

AuthorityEffect: None.
Presiding Tribunal: The Woland Retinue & Lev Davidovich Landau ("Dau").
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Location: Moscow, Sadovaya 302-bis, Apartment 50.
Companion Preface: Read 00_preface_at_apartment_50.md for tribunal staging, character stations, and the Seven Mortal Sins.
Study Guide: Master study_guide.md before sitting for this examination.

> [!IMPORTANT] **Instructions from Professor Landau:** 1. All calculations must be performed in exact rationals $\mathbb{Q} = \{p/q : p \in \mathbb{Z}, q \in \mathbb{Z}_{>0}\}$. 2. Zero floating-point numbers, zero decimals, zero heuristic approximations. 3. An optimization proposal is not a theorem. For this examination, certify every calculated minimum with a feasible dual witness. Complete exact enumeration can also prove a minimum when its search-set theorem and exhaustion are established. 4. Work through the pages in order. Attempt every question and use feedback to repair earlier errors. The complete instructor key has hints, worked answers, and a 100-point rubric; a perfect paper earns the tribunal seal.

Page 1: Horribly Simple (The Kindergarten of Inconsistent Offsets)

Landau tosses his chalk: "If you cannot evaluate the cost of three edges on a napkin, you belong in a business school, not Apartment 50."

Problem 1.1: The Three Inconsistent Clocks

Three instruments report relative time offsets between three nodes a, b, c: - Instrument 1 reports: $s_b - s_a = 1$ (directed edge $e_1 = a \to b$, report $c_1 = 1$). - Instrument 2 reports: $s_c - s_b = 1$ (directed edge $e_2 = b \to c$, report $c_2 = 1$). - Instrument 3 reports: $s_a - s_c = 1$ (directed edge $e_3 = c \to a$, report $c_3 = 1$).

1. Compute the sum of the reports around the directed cycle: $\sum_{i=1}^{3} c_i$.
2. Explain in one sentence why no potential assignment $s = (s_a, s_b, s_c) \in \mathbb{Q}^3$ can satisfy all three reports simultaneously.
3. Does the graph contain any 2-cells (filled faces)? State the dimension of $C^2(G; \mathbb{Q})$.

Problem 1.2: The All-Zero Potential

Let the edge penalties for disagreeing with the reports be $w = (w_1, w_2, w_3) = (1, 2, 3)$. Consider the all-zero candidate potential $s^{(0)} = (0, 0, 0)$. 1. Compute the discrete coboundary vector $Bs^{(0)} \in \mathbb{Q}^3$. 2. Compute the residual vector $r^{(0)} = c - Bs^{(0)}$. 3. Calculate the exact proposal cost:

$$D_c(s^{(0)}) = \sum_{i=1}^{3} w_i |r_i^{(0)}|$$

Problem 1.3: The Candidate Proposal $s = (0, -2, -1)$

A candidate proposes the vertex potential $s^* = (s_a, s_b, s_c) = (0, -2, -1)$. 1. Calculate the potential difference across each edge: - $(Bs^*)_{a\to b} = s_b - s_a =$ _____ - $(Bs^*)_{b\to c} = s_c - s_b =$ _____ - $(Bs^*)_{c\to a} = s_a - s_c =$ _____ 2. Compute the residual 1-cochain $r^* = c - Bs^* = (r_1^*, r_2^*, r_3^*)$. 3. Compute the exact proposal cost $D_c(s^*)$.

Problem 1.4: Proposal vs. Lower Bound Certificate

A machine learning agent runs a search routine, discovers the potential $s^* = (0, -2, -1)$ with cost $D_c(s^*) = 3$, and states: > *"The cost is 3, and since three search heuristics failed to find a lower value, the global minimum is mathematically proved to be 3."*

1. What mathematical bound does $D_c(s^*) = 3$ provide on the quotient minimum $\Phi(c)$?

   Choose: $[\Phi(c) \geq 3 \quad | \quad \Phi(c) \leq 3 \quad | \quad \Phi(c) = 3]$

2. Why does heuristic failure NOT establish that $\Phi(c) = 3$?
3. What mathematical object must be exhibited to certify an independent lower bound $\Phi(c) \geq 3$?

Page 2: Intermediate Rigor (The Primal-Dual Cochain Engine)

Landau taps the slate: "Now we build the dual furnace. Anyone can guess a level; only a mathematician can construct a circulation."

Problem 2.1: Dual Feasibility Conditions

Let $G = (V, E)$ be a directed graph with incidence matrix $B \in \{-1, 0, 1\}^{|E|\times|V|}$, edge reports $c \in \mathbb{Q}^E$, and penalty weights $w \in \mathbb{Q}^E_{>0}$. A vector $\lambda \in \mathbb{Q}^E$ is a candidate dual witness. 1. State the exact linear equation that λ must satisfy

at each vertex $v \in V$ to be a valid circulation. 2. State the inequality constraint that λ_e must satisfy on each edge $e \in E$. 3. If G is a simple directed cycle of length k, show that every valid circulation λ must have constant magnitude across all edges: $\lambda_e = \alpha$ for all $e \in E$.

Problem 2.2: The Weak Duality Derivation

Derive the Weak Duality Theorem: For **every** potential $s \in \mathbb{Q}^V$ and **every** feasible dual circulation $\lambda \in \mathbb{Q}^E$:

$$\langle \lambda, c \rangle \leq D_c(s)$$

Hint: Use $B^T\lambda = \mathbf{0}$ to show that $\sum_e \lambda_e (Bs)_e = 0$, express $\langle \lambda, c \rangle$ in terms of the residual $r = c - Bs$, and apply the capacity bound $|\lambda_e| \leq w_e$. Show every step cleanly.

Problem 2.3: Zero Duality Gap on the Manuscript Triangle

Return to the Manuscript Triangle with $c = (1, 1, 1)$ and $w = (1, 2, 3)$. 1. What is the maximum value that a constant circulation $\lambda = (k, k, k)$ can attain without violating the capacity constraints $|\lambda_i| \leq w_i$? 2. Exhibit the optimal dual witness vector $\lambda^* \in \mathbb{Q}^3$. 3. Compute the dual objective value:

$$\langle \lambda^*, c \rangle = \sum_{i=1}^{3} \lambda_i^* c_i$$

4. Compute the duality gap:

$$\Delta = D_c(s^*) - \langle \lambda^*, c \rangle$$

where $s^* = (0, -2, -1)$. State the definitive conclusion regarding the global minimum $\Phi(c)$.

Problem 2.4: Coboundary Gauge Invariance

Let $c = (1, 1, 1)$ be the reports on the Manuscript Triangle. A gauge transformation $\phi = (2, -1, 5) \in \mathbb{Q}^3$ is applied to the vertices, generating new reports:

$$c' = c + B\phi$$

1. Calculate the shifted report vector $c' \in \mathbb{Q}^3$. 2. Compute the cycle sum $\sum_{i=1}^{3} c'_i$. 3. Without re-running an optimization solver, determine the exact value of $\Phi(c')$. Cite the governing theorem.

Problem 2.5: The Directed Cycle Obstruction Formula

Consider a directed 4-cycle C_4 with vertices $\{v_1, v_2, v_3, v_4\}$, directed edges $e_1 = v_1 \to v_2$, $e_2 = v_2 \to v_3$, $e_3 = v_3 \to v_4$, $e_4 = v_4 \to v_1$. The edge reports are

$c = (4, -2, 5, 1)$ and the penalties are $w = (7, 3, 9, 5)$. 1. State the general Directed Cycle Obstruction Formula for $\Phi(c)$. 2. Identify the bottleneck edge e^* and its weight $\min_e w_e$. 3. Calculate the exact quotient minimum $\Phi(c)$ as an integer or fraction in lowest terms.

Page 3: Adversarial Reality Gates & Invariant Involutions (The Descent Crucible)

Woland smiles faintly. Landau leans in: "Here is where the pretenders drown. If your algorithm descends forever without stopping, Azazello will assist your descent through the floorboards."

Problem 3.1: The Infinite Rational Descent Trap (Zeno's Trap)

Consider a single edge $e = u \to v$ with report $c = 1$ and penalty $w = 1$. An autonomous solver generates a sequence of potential differences $\Delta s_t = s_v(t) - s_u(t)$ such that the residual at step t is $r_t = 2^{-t}$ for $t = 1, 2, 3, \ldots$. 1. Compute the cost D_t at step t. Is the sequence strictly monotonically decreasing ($D_{t+1} < D_t$) for all t? 2. Does there exist any finite step $t^* \in \mathbb{N}$ such that $D_{t^*} = 0$? 3. Explain why the rule *"Admit proposal if $D(s_{new}) < D(s_{old})$"* is an unsafe halting condition in discrete optimization over $\mathbb{Q}$. 4. Separate three obligations: strict `CUT` descent, a finite attempt budget, and a returning or timed-out check. Which are enforced by `SymbolGameGovernor`, which by Demo 1's outer loop, and which by the native adapter? Explain why a turn counter alone is insufficient.

Problem 3.2: Sparsity of Optimal Residuals & The Null-Space Property

Let $G = (V, E)$ be a connected graph with n vertices and m edges ($m \geq n$). 1. State the upper bound on the number of non-zero entries in the residual vector $r^* = c - Bs^*$ for an extreme-point (basic feasible) optimal solution:

$$|\operatorname{supp}(r^*)| \leq _____$$

Express the bound in terms of the cycle rank (first Betti number) $\beta_1 = m - n + 1$. 2. Suppose G is a triangle ($n = 3, m = 3$) where all edge penalties are strictly equal: $w = (2, 2, 2)$, with reports $c = (1, 1, 1)$. - Does there exist a uniquely minimizing potential s^*? - Exhibit two distinct potentials $s^{(1)}$ and $s^{(2)}$ (not differing by an additive constant) that both achieve the global minimum $\Phi(c) = 6$. - Using $f = (0, -2, -1)$ and $\lambda = (2, 2, 2)$, show why a proposed minimum of 2 is impossible. - State the sufficient **Weighted Null-Space Property** using nonzero $z \in \operatorname{im} B$. For $S = \{e_1\}$, exhibit a vector that makes the strict inequality fail. Does failure alone prove that a particular residual is nonunique?

Problem 3.3: Combinatorial Spanning Tree Search & The 100,000 Forest Cap

The native Rust solver `graph_phi1.rs` solves Φ_1 by enumerating spanning forests over G. 1. According to Cayley's formula, the complete graph K_n has n^{n-2} spanning trees. - Calculate the number of spanning trees for K_6. - Calculate the number of spanning trees for K_8. 2. If an examinee passes K_8 to `graph_phi1.rs`, what is the exact behavior of the Rust solver? - Does it return an approximate floating-point answer? - Does it return a partial minimum based on the first 100,000 trees? - What typed enum error does it return? 3. Why does repository governance forbid the solver from returning a "best-so-far" partial minimum when the cap is hit?

Problem 3.4: Lean 4 Toy Involutions and Claim Discipline

In the repository formalization `lean4/`: The manuscript describes the following local statements; inspect the actual files before asserting a build result: - `InvolutionLaws.lean` proves pair cancellation for a supplied sign-reversing, weight-preserving map on signed natural-number weights. - `OddCycleObstruction.lean` concerns the three vertex signs of a triangle. - `NonAuthority.lean` encodes a policy type with one value, so its theorem holds by construction.

An eager machine learning researcher publishes a preprint claiming: > *"The Lean 4 proofs in Paper 001 mathematically verify a polynomial-time algorithm that solves the quantum fermion sign problem."*

Write the formal **Adversarial Receipt Audit** refuting this claim according to the repository Claim Hygiene Rule: 1. **What it supports:** State precisely what `OddCycleObstruction.lean` and `InvolutionLaws.lean` actually formalize in this toy setting. 2. **What it does NOT support:** State three specific physical and computational claims that do NOT follow from these toy Lean files. 3. **The narrowest defensible claim:** Formulate the single narrowest valid claim regarding the Lean 4 formalization in Paper 001. 4. **The next test needed:** Specify the carrier and representation, the mapping theorem, and any additional algorithm or complexity theorem needed for the proposed physical claim. Explain why a toy theorem or a hash receipt cannot supply these obligations.

Grading Trichotomy & Terminal Disposition

Use the problem weights in instructor_key.md. Grade all attempted answers and give a concrete repair for each lost point. These dispositions are fictional game labels, not judgments about a student.

Landau dusts his hands, spins his railway chalk, and glares across the table:

- `PASS` **(Score: 100/100):** Every numerical quantity is in exact rationals $\mathbb{Q}$; all dual circulation witnesses are exhibited with zero duality gap; the weak duality theorem is derived cleanly without missing steps; Cayley's formula and the 100,000 cap are evaluated accurately; and the Lean 4 claim hygiene audit is uncompromisingly strict. *Disposition:* Woland smiles; Landau signs the slate; Jeremy Carroll shakes your hand; Liudmila pours a celebratory cup of tea with strawberry jam.
- `NONDESCENT` **(Score: 40–99/100):** The student confused an upper bound with a certificate, made arithmetic errors on the cycle formula, or handwaved the basic feasible solution sparsity bound. *Disposition:* Koroviev cackles shrilly; the candidate is sent to the kitchen to compute tree enumerations by hand under Polina's supervision.
- `DECAPITATED` **(Score: 0–39/100):** The student used floating-point decimals, asserted that 2^{-t} terminates in finite time, or claimed that toy Lean involutions solve the quantum fermion sign problem. *Disposition:* Azazello points to the open window; Behemoth ignites the primus; Abadonna removes his glasses.

Final challenges in Apartment 50

These optional challenges come after the examination. They do not change its 100-point score. Woland wants a proof; Behemoth wants a counterexample. Hints and solutions are in the instructor key.

Challenge A The sufficient condition that fails

Orient the six edges of K_4 from the earlier letter to the later letter, with vertex order a, b, c, d, and give every edge weight 1. Set $c_{ab} = 1$, $c_{ac} = -1$, and all other reports to 0.

1. Show that $s = 0$ has cost 2. Verify the dual witness, in edge order ab, ac, ad, bc, bd, cd,

$$\lambda = (1, -1, 0, 1/2, 1/2, -1/2).$$

2. Prove that the minimizing potential is unique after fixing $s_a = 0$. Use the strict capacities on the four zero-residual edges.
3. For $S = \{ab, ac\}$ and $h = (1, 0, 0, 0)$, compute $z = Bh$. Show that the sufficient weighted null-space condition fails despite the unique minimizer.
4. State exactly why this does not contradict the theorem in the paper.

Challenge B The perfect-looking counterfeit receipt

The equal-weight triangle has reports $(1, 1, 1)$ and weights $(2, 2, 2)$. A native response claims `PASS`, potential $(0, -2, -1)$, and minimum `2`.

1. Recompute the potential's cost. Which consistency check rejects the response before optimality is considered?

2. Give a feasible dual that excludes a minimum of 2.
3. Write a response-validation checklist covering JSON shape, exact rational strings, complete vertex coverage, forest-count type and range, and objective agreement.
4. Does a structurally valid response and a matching primal cost prove global minimality? Name the remaining trust or certificate obligation. What should happen on a native refusal or timeout?